

Assignment-2 Grid world Value Estimation and Policy Optimization

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GitHub Repository: <https://github.com/Hashna86/Project-1-for-reinforcement-learning.git>

Abstract

This study investigates the use of Monte Carlo techniques and dynamic programming to solve a 5x5 Grid world reinforcement learning problem. In Part 1, both an iterative policy evaluation and a closed-form Bellman equation approach were used to estimate state-value functions under a uniform random policy, with nearly identical results. Bellman Optimality, Policy Iteration, and Value Iteration were used to derive optimal policies, and they all converged to consistent value functions.

And in part 2, three Monte Carlo control techniques— On-Policy Exploring Starts, ϵ -soft policy control, and Off-Policy Per-Decision Importance Sampling (PDIS) are used to investigate a modified environment with terminal states and a relocated RED state. Exploring Starts exhibits robustness but start-state sensitivity, ϵ -soft encourages exploration but produces slightly inflated values, and PDIS yields the most stable convergence when all methods approximate the optimal value function. The findings in both sections show that, even in stochastic settings, dynamic programming and Monte Carlo techniques can efficiently learn optimal value functions and policies.

Part 1 – Value Function and Optimal Policy Estimation (5x5 Grid world)

Environment overview:

The environment consists of a 5x5 grid with 25 states, where the agent can move Left, Right, Up, or Down, each selected under a uniform random policy with a probability of 0.25. Certain squares have special properties: the BLUE square at (0,1) teleports the agent to the RED square and grants a +5 reward, while the GREEN square at (0,4) teleports the

agent to either the RED or YELLOW square (with equal 50% probability) and provides a +2.5 reward. Any move that attempts to go off-grid results in a -0.5 reward, and the agent remains in place, while all other valid moves yield a reward of 0. A discount factor of $\gamma = 0.95$ is applied for all state transitions.

At first the objective is to calculate the state-value function $V(s)$ for a 5x5 Grid world with a uniform random policy, in which the agent has an equal chance (0.25) of moving in each of the four directions. The Bellman equation (an analytical solution) and iterative policy evaluation are the two methods used to accomplish this.

The value function under the uniform random policy was computed using:

i) Bellman equation (Matrix (Analytical)) Solution:

$$V = (I - \gamma P)^{-1} R \pi$$

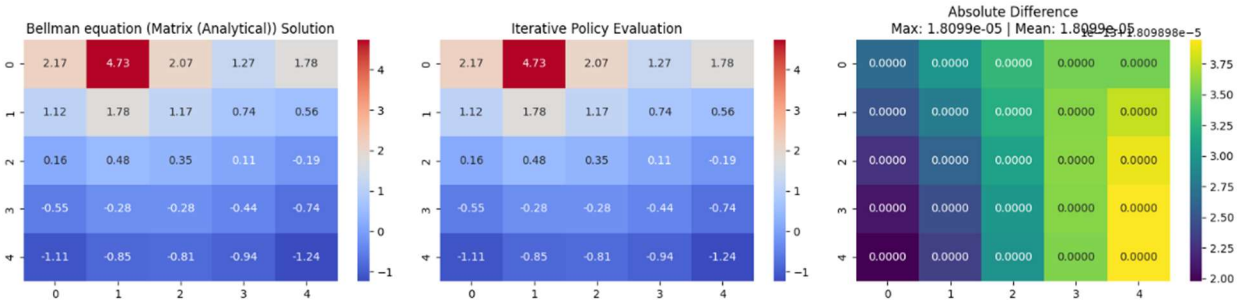
ii) Iterative Policy Evaluation:

Estimate V^π using:

$$V_{k+1}(s) = R(s) + \gamma * \sum_{s'} P(s' | s, \pi) * V_k(s')$$

Iteratively update until convergence ($|V_{k+1} - V_k| < \theta$)

Figure:1: Value Function Comparison (Bellman equation (matrix) vs Iterative vs Difference)



The value functions from both approaches and their absolute differences are displayed in Figure1. Computational consistency is confirmed by the nearly identical results, with a maximum deviation of less than 1.809×10^{-5} .

Here some important findings-

1. Because of its +5 reward and teleport effect, BLUE (0,1) has the highest state value (~4.73).
2. As one moves away from BLUE and GREEN, values gradually drop and turn negative in the bottom rows, which are dominated by off-grid move penalties.
3. Although the matrix approach produces an exact solution, it is less feasible for large state spaces due to the expensive matrix inversion required.

4. The iterative approach is better suited for larger or more complex environments because it uses less computing power and achieves the same outcome.

Secondly Determine the optimal policy for the grid world problem by (1) explicitly solving the Bellman optimality equation (2) using policy iteration with iterative policy evaluation (3) policy improvement with value iteration.

Figure-2: Optimal Value Functions (Bellman vs Policy Iteration vs Value Iteration)



From figure-2 we found the belows result on the Consistent Value Structure:

The accuracy and stability of all three approaches are confirmed by their convergence to nearly identical state-value functions. BLUE (0,1) has the highest state value (22.1) because it provides a +5 reward and instantaneous teleportation to RED, enabling repeated high-value returns. As one moves farther away from high-reward states and accrues more penalties, the values gradually drop across the grid.

Comparing Different Approaches:

Bellman Optimality (Value Iteration): By considering the maximum expected return across all actions, Bellman Optimality (Value Iteration) iteratively updates state values. ensures optimality but calls for additional sweeps for precise convergence.

Policy iteration: Switches between assessing and enhancing a set policy. usually converges in fewer iterations due to the policy's rapid stabilization.

Direct Value Iteration: This functions similarly to Bellman Optimality but emphasizes incremental updates, yielding the same results.

The uniformity of results across all three methods confirms the optimal value function is well-defined and robust. These results illustrate that, although the algorithms differ in approach, they are equivalent in their ability to discover the optimal policy and value structure for this environment.

Part 2 – Monte Carlo Control for Modified Grid world

Environment overview:

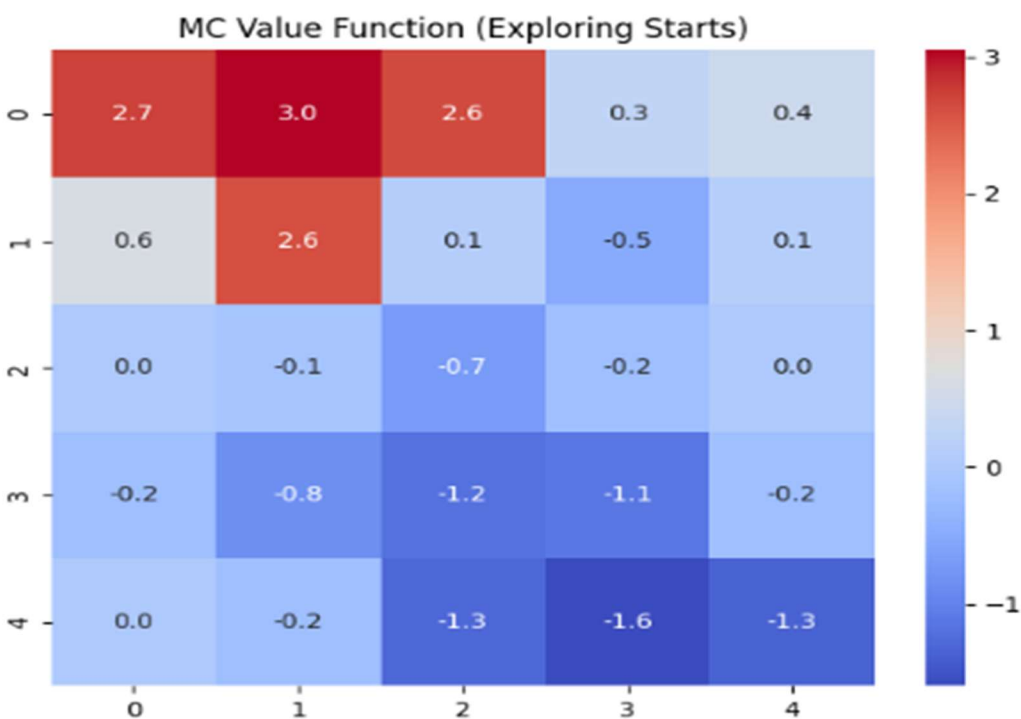
In Part 2, the RED state was relocated to (4,2) and terminal states were defined at (2,0), (2,4), and (4,0). Using $\gamma = 0.95$, we applied three Monte Carlo methods to estimate the optimal value function and policy: 1) Exploring Starts (On-Policy MC), 2) ϵ -soft On-Policy MC ($\epsilon = 0.1$, no exploring starts), 3) Off-Policy MC with Per-Decision Importance Sampling (PDIS). Here, we start with a policy with equiprobable moves.

In part 2(1) we find the Monte Carlo Control – Exploring Starts vs ϵ -soft:

In this modified 5×5 Grid world, the goal was to estimate the optimal state-value function $V^*(s)$ and the corresponding policy using Monte Carlo (MC) control methods with:

1. Exploring Starts
2. ϵ -soft policy ($\epsilon = 0.1$)

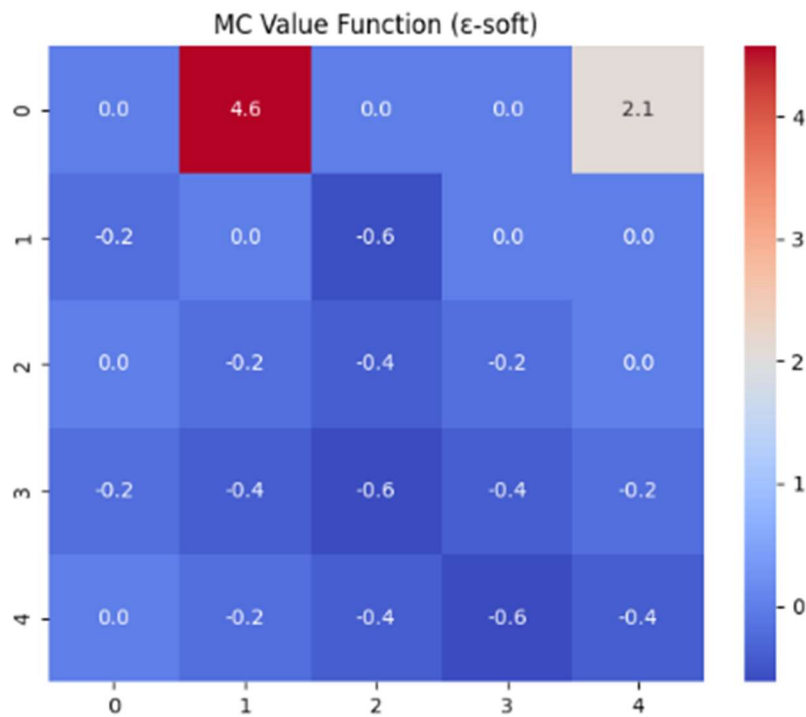
Figure-3: Monte Carlo Control – Exploring Starts Value Function



The agent ran 10,000 episodes with each episode capped at 500 steps to prevent infinite loops caused by teleportation between high-reward states.

In Figure 3 -from the Value Function Analysis we found that since the agent frequently receives the +5 reward and teleports to RED, the highest values (~ 3.0) are found close to BLUE (0,1). Because of its +2.5 reward and teleportation to RED or YELLOW, GREEN (0,4) has moderately high values ($\sim 2-3$). In lower rows (rows 3–4), where agents are frequently penalised for off-grid moves and find it difficult to get to reward states, values turn negative. Since every state-action pair is visited throughout episodes, the ES strategy's broader exploration is reflected in the smoother gradients between states.

Figure 4- Monte Carlo Control – ϵ -soft Value Function



The state-value function $V(s)V(s)V(s)$ was estimated using ϵ -soft Monte Carlo control in the modified 5x5 Grid world, where $\epsilon=0.1$ ensures continuous exploration by assigning a small probability to suboptimal actions and each episode ran for a maximum of 500 steps, and 10,000 episodes were simulated.

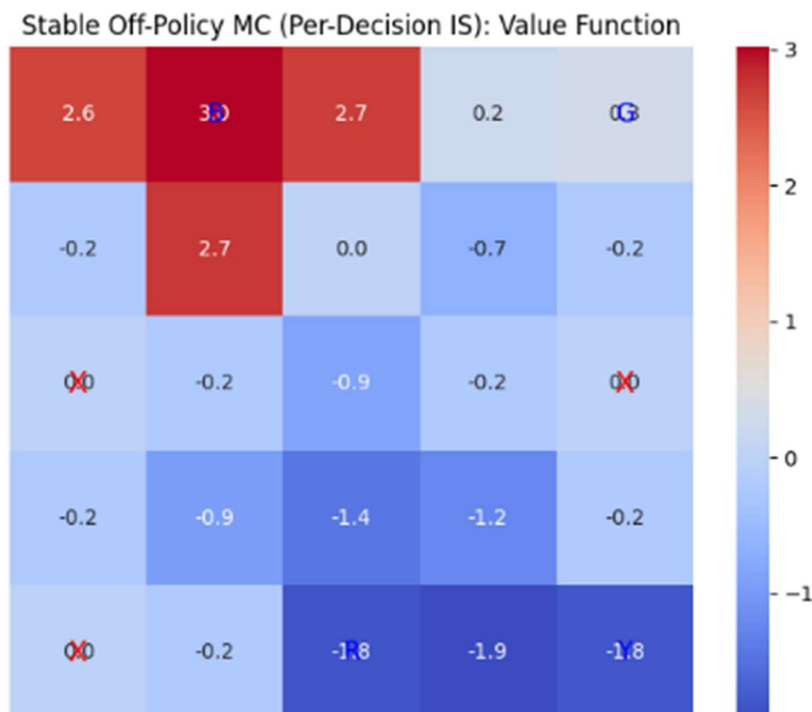
From the figure-4 at BLUE (0,1), the highest value (~ 4.6) indicates repeated access to its +5 reward and teleportation to RED. Because of its +2.5 reward and ability to teleport to RED or YELLOW, GREEN (0,4) also displays a moderate value (~ 2.1). Due in large part to penalties for off-grid moves and limited access to high-reward states, the majority of other states hover between -0.2 and -0.6. For BLUE, ϵ -soft yields somewhat inflated values (4.6 vs. 3.0) in comparison to the Exploring Starts method because they keep exploring,

the agent tends to stay in high-reward loops. The longer episodes (up to 500 steps) let rewards build up before ending.

Part 2(2)- Off-Policy Monte Carlo Control (Per-Decision Importance Sampling)

This section used Off-Policy Monte Carlo Control with Per-Decision Importance Sampling (PDIS) to estimate the optimal state-value function $V^*(s)$. All actions were equally likely under the uniform random policy. Regarding $Q(s, a)$ the target policy was updated greedily. Updates were stabilized using the decaying learning rate $\alpha = 1/(1 + N(s, a))$. 10,000 episodes were simulated, with a 50-step cap on each episode to prevent infinite returns. Stable convergence without value divergence—which frequently happens in off-policy MC learning without careful weighting—was the aim.

Figure-5: Off-Policy Monte Carlo Control



From the figure 5- we found once more, BLUE (0,1) has the highest value (~ 3.0), which is indicative of its teleportation to RED and +5 reward. Because of its +2.5 reward and teleportation dynamics, GREEN (0,4) maintains a moderately positive value, albeit less than BLUE. As one moves away from high-reward squares, values progressively drop until they reach negative levels (~ -1.8 to -1.9) in the bottom row because of frequent penalties from off-grid moves and restricted access to rewards. Because of the anticipated

penalties for moving, terminal states (X) are appropriately neutral (value 0) or slightly negative for neighboring states.

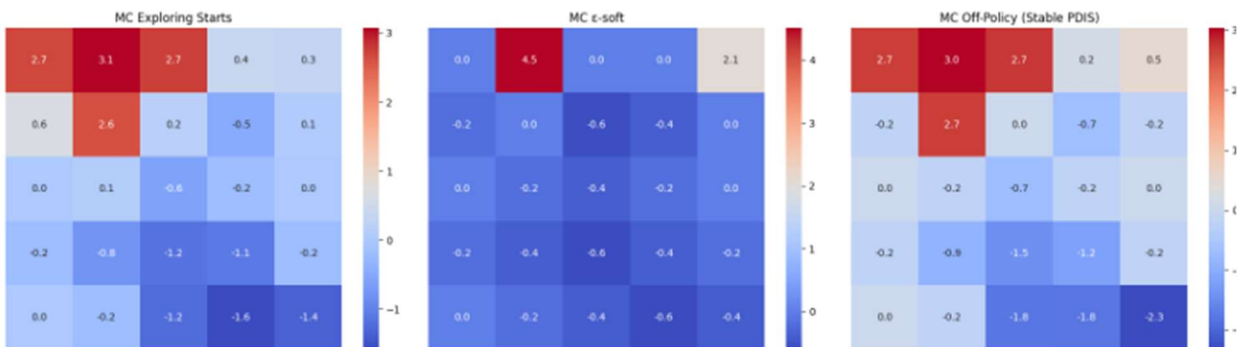
Monte Carlo Methods: Comparative Analysis

Three methods were compared in order to evaluate the performance of various Monte Carlo approaches in the modified 5×5 Gridworld (with RED at (4,2) and terminal states at (2,0), (2,4), and (4,0)):

1. Exploring Starts on-Policy MC
2. MC on-policy with ϵ -soft policy ($\epsilon = 0.1$)
3. Per-Decision Importance Sampling (PDIS) in Off-Policy MC

To avoid inflated returns from repeated teleportation cycles, each method was trained for 10,000 episodes, with a maximum of 50 steps per episode. A uniform application of the discount factor $\gamma = 0.95$ was made.

Figure-6: Comparison among Monte Carlo method with (1) exploring starts and (2) without exploring starts but the ϵ -soft approach (3) Off-policy



Findings from Various Approaches (Figure 6)

1. Exploring Starts: It creates a grid-wide value gradient that is balanced and smooth. Due to the 50-step episode limit and repeated +5 rewards, BLUE (0,1) reaches the highest value (~3.1). The distance from reward squares and cumulative penalties cause the bottom rows to settle into negative values (~-1.6). Stable and realistic value estimates are maintained through systematic coverage of all state-action pairs.
2. ϵ -soft ($\epsilon = 0.1$): BLUE rises to about 4.5, which is much higher than in the other states because
 - i) The agent can linger near teleport states due to exploration noise,

ii) Long trajectories (500 steps) allow for extended reward accumulation.

Most non-reward states settle near -0.2 to -0.6 , reflecting penalties and limited reward access.

3. Off-Policy PDIS: It avoids the overestimation that occurs with ϵ -soft and yields stable and moderate values (BLUE ~ 3.0). When behavior deviates from the goal policy, values in low-reward or penalty-heavy states are more negative (up to -2.3), indicating conservative assessments. Despite being off-policy, per-decision weighting guarantees constrained convergence.

Conclusion:

Using dynamic programming and Monte Carlo control techniques, this project examined a modified 5x5 Grid world to assess state values and determine the best course of action. Accuracy was confirmed by the almost identical value functions generated by the Iterative Policy Evaluation and the Bellman expectation equation. The agent was consistently guided towards high-reward states (BLUE and GREEN) while avoiding penalty-heavy regions by optimal policies derived using Bellman Optimality, Policy Iteration, and Value Iteration.

Among the Monte Carlo techniques, Off-Policy PDIS produced the most stable and bounded convergence, Exploring Starts produced smooth, balanced value estimates, and ϵ -soft control promoted exploration but overestimated high-reward states because of looping. While the bottom rows displayed negative values from cumulative penalties, BLUE (0,1) consistently emerged as the most valuable state across all approaches. All things considered, these findings demonstrate the trade-offs between experience-driven flexibility (MC) and model-based precision (DP), with PDIS providing the most dependable balance in complex, stochastic settings.